Valerio Di Zio

Cloud Developer / Cloud Engineer

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Professional profile

I'm Valerio, a cloud developer. I currently work on Google Cloud projects: I build backend services, mainly in Python but not exclusively, and work directly with clients to understand what they need and how best to approach it.

I graduated with honours in 2025 from the University of Bologna (Cesena campus), with an MSc in Computer Science and Engineering and a thesis on fairness in machine learning. I'm mostly interested in cloud architecture and artificial intelligence.

Outside work, I like cooking, 3D printing, video games and collecting. I try to stay active and spend time with friends.

Education

MSc in Computer Science and Engineering

University of Bologna, Cesena | 2022 - 2025 | **Final grade: 110/110 cum laude** (highest distinction)

Professional experience

Cloud Developer / IT Consultant

Go Reply | September 2025 - present

- Contribute to Google Cloud solution design and development, from requirements gathering to POC evaluation and architecture decisions.
- Build Python and FastAPI services for serverless applications, data processing and cloud integrations.
- Develop Google ADK agents integrated with BigQuery; contribute to CI/CD, IAM, monitoring, logging, troubleshooting and technical support.

IT Trainer

SER.IN.AR | 2024 - 2025

Delivered four courses to four groups: basic Excel for school staff, LEGO Mindstorms robotics for a middle school class, and Snap programming for two primary school classes.

Frontend Developer

Visup | May 2023 - December 2023

Developed the Angular frontend of an Industry 4.0 data visualisation dashboard, collaborating in an Agile team with Git and CI/CD.

Certifications

Certified Partner Specialist Gemini Enterprise Deployment
Google Cloud | September 2026 - March 2027 | View credential

Generative AI Leader Certification
Google Cloud | November 2025 - November 2028 | View credential

Claude 101
Certificate of completion | Anthropic | April 2026 | Credential ID: s8t7chuqcvas

Professional Cloud Developer
Certification in preparation | Target: late October 2026

Technical skills

Technologies used at work and in university and personal projects.

Languages: Python, Kotlin, Scala, Java, TypeScript, JavaScript, SQL.

Backend & integrations: FastAPI, Pydantic, Node.js, Express, gRPC, RabbitMQ, Socket.IO, Akka Typed.

Frontend: Angular (previous experience), Vue.js, React, HTML, CSS.

Data: PostgreSQL, MongoDB, NumPy, pandas, SciPy, Apache Spark / MLlib, Amazon S3.

AI & machine learning: Google ADK, Datapizza AI, scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, MidiTok; fairness and reinforcement learning.

Google Cloud (GCP): Cloud Run, Cloud Storage, BigQuery, Firestore, Pub/Sub, Cloud Tasks, Cloud Scheduler, IAM, Secret Manager, Cloud Monitoring, Cloud Logging.

DevOps & build: Git, Docker, Docker Compose, GitHub Actions, Cloud Build, CI/CD, Gradle, sbt, Poetry.

Testing: pytest, unittest, ScalaTest, Kotest, JUnit, Cucumber (BDD), Playwright.

Architecture & methods: Serverless, microservices, hexagonal architecture, event-driven systems, functional programming, DSLs, Agile/Scrum.

Selected projects

FairLib, MSc thesis

Python, machine learning fairness

Python library implementing pre-processing and in-processing fairness techniques, used in the European AEQUITAS project. Listed as a contributor to an AEQUITAS research dataset on Zenodo.

GitHub repository | Dataset on Zenodo

PositionPal, User Service

Kotlin, gRPC, RabbitMQ, PostgreSQL

User registration and profile microservice with hexagonal architecture, automated tests and contributions to the CI/CD pipeline.

GitHub repository | Project documentation

NesGen, generative music

Python, TensorFlow, PyTorch, MidiTok

Transformer and GAN approaches for generating NES-style MIDI music, including tokenisation and fine-tuning on the NES-MDB dataset.

GitHub repository

Scooby, web crawler

Scala 3, Akka Typed, internal DSL

Crawler and scraper with a DSL for navigation, extraction and export; Akka Typed manages concurrent execution.

GitHub repository

Languages

Italian: native | **English:** B2, university exam

Available for remote, hybrid or on-site opportunities.